Atiyah–Macdonald, Chapter 11: Dimension Theory

Solutions to the exercises

Conventions. Rings are commutative with 1; ht denotes height of a prime ideal and dim the Krull dimension. From the text I use: Krull's principal ideal theorem in the form 11.16 (in a Noetherian ring every minimal prime ideal of $(x_1, \dots, x_r)$ has height $\leq r$); 11.12 (heights are finite in a Noetherian ring); 11.13 (a Noetherian local ring of dimension d has an $\mathfrak{m}$-primary ideal generated by d elements); 11.14 (dimension theorem), together with the example after it ($\dim k[x_1, \dots, x_n]_{(x_1,\dots,x_n)} = n$); 11.18 ($\dim A/(x) = \dim A - 1$ for a non-zero-divisor $x \in \mathfrak{m}$); 11.20 and 11.21; 11.22 (regularity: $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim A$); 11.26 (integral extension of integrally closed domains preserves local dimension). Also the Normalization Lemma (Ch. 5, Ex. 16), 5.8–5.10 (maximal ideals, lying over), 10.24 (finite generation of complete modules), Ch. 3 Ex. 21 and Ch. 4 Ex. 7 (the ideals $\mathfrak{p}[x]$).

This chapter has no numbered in-text exercises; there are 7 end-of-chapter exercises. Exercise 4 (Nagata) asks us to verify several assertions made in its statement; each is proved below. Exercise 5 refers to the Grothendieck group $K(A_0)$ of Ch. 7, Ex. 26, which I recall in the statement.

End-of-chapter exercises

Exercise 1

> **Problem.** Let $f \in k[x_1, \dots, x_n]$ be an irreducible polynomial over an algebraically closed field k. A point P on the variety $f(x) = 0$ is *non-singular* $\iff$ not all the partial derivatives $\partial f/\partial x_i$ vanish at P. Let $A = k[x_1, \dots, x_n]/(f)$, and let $\mathfrak{m}$ be the maximal ideal of A corresponding to the point P. Prove that P is non-singular $\iff$ $A_\mathfrak{m}$ is a regular local ring. [By (11.18) we have $\dim A_\mathfrak{m} = n - 1$. Now $\mathfrak{m}/\mathfrak{m}^2 \cong (x_1, \dots, x_n)/\big((x_1, \dots, x_n)^2 + (f)\big)$ and has dimension $n - 1$ if and only if $f \notin (x_1, \dots, x_n)^2$.]

Solution. Write $P = (a_1, \dots, a_n)$, $B = k[x_1, \dots, x_n]$, $M = (x_1 - a_1, \dots, x_n - a_n)$ (the maximal ideal of P in B). Since P lies on the variety, $f \in M$, and $f \neq 0$. The ring $A = B/(f)$ is a domain (f irreducible, B a UFD), and $\mathfrak{m} = M/(f)$.

Dimension. B_M is a Noetherian local domain of dimension n (translate P to the origin and use the example after 11.14). $f \in MB_M$ is a non-zero element of a domain, hence a non-zero-divisor in the maximal ideal. By 11.18, $\dim A_\mathfrak{m} = \dim B_M/(f) = n - 1$.

Cotangent space. $\mathfrak{m}/\mathfrak{m}^2$ is a vector space over $k = A/\mathfrak{m}$ killed by $\mathfrak{m}$, so it is unchanged by localization at $\mathfrak{m}$: $\mathfrak{m}A_\mathfrak{m}/\mathfrak{m}^2A_\mathfrak{m} \cong \mathfrak{m}/\mathfrak{m}^2$. Moreover $\mathfrak{m}^2 = (M^2 + (f))/(f)$, so

$$\mathfrak{m}/\mathfrak{m}^2 \cong M/(M^2 + (f)) = \big(M/M^2\big)\big/\big(\text{image of } f\big).$$

M/M^2 has k-basis the classes of $x_i - a_i$ $(1 \le i \le n)$, so $\dim_k M/M^2 = n$. Expand f in powers of $y_i = x_i - a_i$ (Taylor expansion at P, valid for formal derivatives in every characteristic): $f = f(P) + \sum_i \frac{\partial f}{\partial x_i}(P)\, y_i + (\text{terms in } M^2)$, with $f(P) = 0$. Hence the image of f in M/M^2 is $\sum_i \frac{\partial f}{\partial x_i}(P)\, \bar{y}_i$, which is non-zero iff some partial derivative does not vanish at P. So

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = \begin{cases} n-1 & \text{if } P \text{ is non-singular,} \\ n & \text{if } P \text{ is singular.} \end{cases}$$

By 11.22, $A_\mathfrak{m}$ is regular iff $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim A_\mathfrak{m} = n - 1$, i.e. iff P is non-singular. □

Exercise 2

> **Problem.** In (11.21) assume that A is complete. Prove that the homomorphism $k[[t_1, \dots, t_d]] \to A$ given by $t_i \mapsto x_i$ $(1 \le i \le d)$ is injective and that A is a finitely-generated module over $k[[t_1, \dots, t_d]]$. [Use (10.24).]

Solution. Recall the setting of 11.21: A is a Noetherian local ring with maximal ideal $\mathfrak{m}$, $k \subseteq A$ a field mapping isomorphically onto $A/\mathfrak{m}$, $x_1, \dots, x_d$ a system of parameters ($d = \dim A$), and $\mathfrak{q} = (x_1, \dots, x_d)$ is $\mathfrak{m}$-primary. Put $B = k[[t_1, \dots, t_d]]$.

The homomorphism. Since $\mathfrak{m}^r \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ for some r, the $\mathfrak{q}$-adic and $\mathfrak{m}$-adic topologies on A agree, and A is complete for them. For $g = \sum c_\alpha t^\alpha \in B$ $(c_\alpha \in k)$ the series $\sum c_\alpha x^\alpha$ converges in A $(x^\alpha \in \mathfrak{q}^{|\alpha|})$, and $\phi(g) = \sum c_\alpha x^\alpha$ defines a k-algebra homomorphism $\phi : B \to A$ with $\phi((t)^n) \subseteq \mathfrak{q}^n$, where $(t) = (t_1, \dots, t_d)$.

Step 1: $G_{(t)}(B) = k[t_1, \dots, t_d] \to G_\mathfrak{q}(A)$, $t_i \mapsto \bar{x}_i$, is injective and makes $G_\mathfrak{q}(A)$ a finitely generated module. This is the map induced by ϕ on associated graded rings. *Injective:* the map is graded, so it suffices to show that no non-zero homogeneous $h \in k[t]$ of degree s is in its kernel, i.e. that $h(x) \in \mathfrak{q}^{s+1}$ forces $h = 0$. By 11.20, if $h(x) \in \mathfrak{q}^{s+1}$ then all coefficients of h lie in $\mathfrak{m}$; but they lie in k and $k \cap \mathfrak{m} = 0$ ($k \to A/\mathfrak{m}$ is injective). So $h = 0$. *Finite generation:* $A/\mathfrak{q}$ has finite length (it is Artin), so it is a finite-dimensional k-vector space (every composition factor is $A/\mathfrak{m} \cong k$), say spanned by $e_1, \dots, e_N$. $G_\mathfrak{q}(A)_n = \mathfrak{q}^n/\mathfrak{q}^{n+1}$ is spanned over $A/\mathfrak{q}$ by the classes of the monomials of degree n in x_i, hence over k by $e_j \cdot \bar{x}^\alpha$. So the $e_j \in G_\mathfrak{q}(A)_0$ generate $G_\mathfrak{q}(A)$ as a $k[t]$-module.

Finite generation of A over B. Regard A as a B-module via ϕ with the filtration $A_n = \mathfrak{q}^n$; then $(t)A_n \subseteq A_{n+1}$, so this is a (t)-filtration. B is complete in the (t)-adic topology, A is Hausdorff in the filtration topology ($\bigcap \mathfrak{q}^n = 0$ by Krull's theorem, 10.17, as A is Noetherian local), and $G(A)$ is a finitely generated $G(B)$-module by Step 1. By 10.24, A is a finitely generated B-module.

Injectivity of ϕ. Let $0 \ne g \in B$ with $\phi(g) = 0$. Write $g = g_s + g_{s+1} + \cdots$ with g_j homogeneous of degree j and $g_s \ne 0$. Every monomial x^α with $|\alpha| > s$ can be written $x^\beta x^{\alpha-\beta}$ with $|\beta| = s+1$ (choose β for each α), so grouping the convergent series $\sum_{|\alpha|>s} c_\alpha x^\alpha = \sum_{|\beta|=s+1} x^\beta u_\beta$ with

$u_\beta \in A$; this lies in $\mathfrak{q}^{s+1}$. Then $0 = \phi(g) = g_s(x) + (\text{element of } \mathfrak{q}^{s+1})$ gives $g_s(x) \in \mathfrak{q}^{s+1}$, hence $g_s = 0$ by Step 1 (again 11.20), a contradiction. So ϕ is injective. □

Exercise 3

> **Problem.** Extend (11.25) to non-algebraically-closed fields. [If $\bar{k}$ is the algebraic closure of k, then $\bar{k}[x_1, \ldots, x_n]$ is integral over $k[x_1, \ldots, x_n]$.]

Solution. *Statement.* Let k be *any* field, $\mathfrak{p}$ a prime ideal of $k[x_1, \ldots, x_n]$, $A = k[x_1, \ldots, x_n]/\mathfrak{p}$, K its field of fractions and $d = \operatorname{tr.deg}_k K$. Then $\dim A_\mathfrak{m} = d$ for every maximal ideal $\mathfrak{m}$ of A; consequently $\dim A = d$. (For k algebraically closed, maximal ideals are the points of the variety, and this is 11.25 and 11.27.)

Step 1: for $B = k[y_1, \ldots, y_d]$ (independent indeterminates) and every maximal ideal $\mathfrak{n}$ of B, $\dim B_\mathfrak{n} = d$. Let $\bar{k}$ be an algebraic closure of k and $\bar{B} = \bar{k}[y_1, \ldots, y_d] \supseteq B$. Every element of $\bar{k}$ is algebraic over k, hence integral over B; together with the $y_i \in B$ this shows $\bar{B}$ is integral over B (the integral closure of B in $\bar{B}$ is a subring containing $\bar{k}$ and B). B is a UFD, hence integrally closed (5.12). By lying over (5.10) there is a prime $\bar{\mathfrak{m}}$ of $\bar{B}$ with $\bar{\mathfrak{m}} \cap B = \mathfrak{n}$, and $\bar{\mathfrak{m}}$ is maximal by 5.8. By Lemma 11.26, $\dim B_\mathfrak{n} = \dim \bar{B}_{\bar{\mathfrak{m}}}$. As $\bar{k}$ is algebraically closed, $\bar{\mathfrak{m}} = (y_1 - c_1, \ldots, y_d - c_d)$ for some $c_i \in \bar{k}$ (Nullstellensatz, 7.9), and after translating c to the origin the example after 11.14 gives $\dim \bar{B}_{\bar{\mathfrak{m}}} = d$.

Step 2: the general case. By the Normalization Lemma (Ch. 5, Ex. 16; the substitution $x_i \mapsto x_i - x_n^{N^i}$ works over any field) there are $y_1, \ldots, y_r \in A$ algebraically independent over k with A integral over $B = k[y_1, \ldots, y_r]$. Then K is algebraic over $k(y_1, \ldots, y_r)$, so $r = \operatorname{tr.deg}_k K = d$. B is integrally closed (UFD) and A is an integral domain integral over B. For a maximal ideal $\mathfrak{m}$ of A, $\mathfrak{n} = \mathfrak{m} \cap B$ is maximal (5.8) and Lemma 11.26 gives $\dim A_\mathfrak{m} = \dim B_\mathfrak{n} = d$ by Step 1. Hence $\dim A = \sup_\mathfrak{m} \dim A_\mathfrak{m} = d$ too. □

Exercise 4

> **Problem.** (An example of a Noetherian domain of infinite dimension, Nagata.) Let k be a field and let $A = k[x_1, x_2, \ldots, x_n, \ldots]$ be a polynomial ring over k in a countably infinite set of indeterminates. Let $m_1, m_2, \ldots$ be an increasing sequence of positive integers such that $m_{i+1} - m_i > m_i - m_{i-1}$ for all $i > 1$. Let $\mathfrak{p}_i = (x_{m_i+1}, \ldots, x_{m_{i+1}})$ and let S be the complement in A of the union of the ideals $\mathfrak{p}_i$. Each $\mathfrak{p}_i$ is a prime ideal and therefore the set S is multiplicatively closed. The ring $S^{-1}A$ is Noetherian by Chapter 7, Exercise 9. Each $S^{-1}\mathfrak{p}_i$ has height equal to $m_{i+1} - m_i$, hence $\dim S^{-1}A = \infty$.

Solution. Put $r_i = m_{i+1} - m_i$. Call $X_i = \{x_{m_i+1}, \ldots, x_{m_{i+1}}\}$ the i-th block of variables; the blocks are pairwise disjoint.

$\mathfrak{p}_i$ is prime and S is multiplicatively closed. $A/\mathfrak{p}_i$ is the polynomial ring in all variables outside X_i, a domain. The complement of a union of primes is multiplicatively closed ($1 \in S$). A is a domain, so $S^{-1}A$ is a domain.

(a) A non-zero $a \in A$ lies in only finitely many $\mathfrak{p}_i$. A polynomial involves finitely many variables. If no variable of X_i occurs in a, then $a \in \mathfrak{p}_i$ would force $a = 0$ (a polynomial lies in $\mathfrak{p}_i = (X_i)$ iff each of its monomials contains a variable of X_i).

(b) Every prime $\mathfrak{q}$ of A with $\mathfrak{q} \cap S = \emptyset$ (i.e. $\mathfrak{q} \subseteq \bigcup \mathfrak{p}_i$) is contained in some $\mathfrak{p}_i$. Suppose not; we construct an element of $\mathfrak{q}$ outside every $\mathfrak{p}_j$, contradicting $\mathfrak{q} \subseteq \bigcup \mathfrak{p}_j$. Clearly $\mathfrak{q} \neq 0$ (as $0 \subseteq \mathfrak{p}_1$); take $0 \neq x \in \mathfrak{q}$ and let $I = \{i : x \in \mathfrak{p}_i\}$, finite by (a); $I \neq \emptyset$ since $x \notin S$. As $\mathfrak{q} \not\subseteq \mathfrak{p}_i$ for $i \in I$, prime avoidance (1.11) gives $z \in \mathfrak{q}$ with $z \notin \mathfrak{p}_i$ for all $i \in I$. Choose a variable x_N not occurring in x and not belonging to any block X_i with $i \in I$ (possible: infinitely many variables). Put $u = x + zx_N \in \mathfrak{q}$.

- For $i \in I$: $x \in \mathfrak{p}_i$ while $z, x_N \notin \mathfrak{p}_i$, so $zx_N \notin \mathfrak{p}_i$ (prime) and $u \notin \mathfrak{p}_i$.
- For $j \notin I$: $x \notin \mathfrak{p}_j$, so x has a monomial μ containing no variable of X_j. No monomial of zx_N equals μ, because every monomial of zx_N is divisible by x_N while μ is not. So μ occurs in u with its coefficient unchanged, and μ has no variable of X_j; hence $u \notin \mathfrak{p}_j$.

Thus $u \in \mathfrak{q} \cap S$, a contradiction. So $\mathfrak{q} \subseteq \mathfrak{p}_i$ for some i.

(c) Maximal ideals of $R = S^{-1}A$. Primes of R are $S^{-1}\mathfrak{q}$ with $\mathfrak{q}$ a prime of A disjoint from S (3.11), inclusion-preserving. By (b) each such $\mathfrak{q}$ lies in some $\mathfrak{p}_i$; and $\mathfrak{p}_i \subseteq \mathfrak{p}_j$ forces $i = j$ (the blocks are disjoint, so $x \in X_i$ is not in $\mathfrak{p}_j$ for $j \neq i$). Hence the maximal ideals of R are exactly the $\mathfrak{n}_i = S^{-1}\mathfrak{p}_i$.

(d) $R_{\mathfrak{n}_i}$ is Noetherian and $\dim R_{\mathfrak{n}_i} = r_i$. Since $S \subseteq A \setminus \mathfrak{p}_i$, $R_{\mathfrak{n}_i} \cong A_{\mathfrak{p}_i}$ (Ch. 3, Ex. 3). Let $T = k[x_j : x_j \notin X_i] \setminus \{0\} \subseteq A \setminus \mathfrak{p}_i$ and let F be the field of fractions of $k[x_j : x_j \notin X_i]$. Then $T^{-1}A = F[X_i]$ is the polynomial ring in r_i variables over F, and $A_{\mathfrak{p}_i} = (T^{-1}A)_{T^{-1}\mathfrak{p}_i} = F[X_i]_{(X_i)}$, the localization of a polynomial ring at the ideal generated by all its variables. This is a Noetherian local ring of dimension r_i (example after 11.14).

(e) R is Noetherian. By Ch. 7, Ex. 9 (a ring in which every localization at a maximal ideal is Noetherian and every non-zero element lies in only finitely many maximal ideals is Noetherian) it suffices to check these two conditions. The first is (d). For the second let $0 \neq a/s \in R$; then $a \neq 0$ and $a/s \in \mathfrak{n}_i \iff a \in \mathfrak{p}_i$ (as $\mathfrak{p}_i \cap S = \emptyset$), which holds for only finitely many i by (a).

(f) Dimension. $\operatorname{ht} \mathfrak{n}_i = \dim R_{\mathfrak{n}_i} = r_i = m_{i+1} - m_i$ by (d). The hypothesis $m_{i+1} - m_i > m_i - m_{i-1}$ says the r_i form a strictly increasing sequence of integers, so $r_i \to \infty$. Therefore $\dim S^{-1}A \geq \sup_i r_i = \infty$: $S^{-1}A$ is a Noetherian domain of infinite Krull dimension. (Each local ring $R_{\mathfrak{n}_i}$ has finite dimension, in agreement with Cor. 11.11, but these dimensions are unbounded.) $\square$

Exercise 5

> **Problem.** Reformulate (11.1) in terms of the Grothendieck group $K(A_0)$ (Chapter 7, Exercise 26).
>
> (Recall: for a Noetherian ring A_0, $K(A_0)$ is the quotient of the free abelian group on isomorphism classes (M) of finitely generated A_0-modules by the subgroup generated by $(M') - (M) + (M'')$ for all short exact sequences $0 \to M' \to M \to M'' \to 0$; $\gamma(M)$ denotes the class of M. By Ch. 7, Ex. 26(i), γ is the *universal* additive function: for every additive function λ on finitely generated A_0-modules with values in an abelian group G there is a unique homomorphism $\lambda_0 : K(A_0) \to G$ with $\lambda = \lambda_0 \circ \gamma$.)

Solution. *Reformulation.* Let $A = \bigoplus_{n \geq 0} A_n$ be a Noetherian graded ring (so A_0 is Noetherian by 10.7), generated over A_0 by homogeneous elements $x_1, \ldots, x_s$ of degrees $k_1, \ldots, k_s > 0$, and

let $M = \bigoplus M_n$ be a finitely generated graded A-module. Each M_n is a finitely generated A_0-module, so its class $\gamma(M_n) \in K(A_0)$ is defined. Define the *universal Poincaré series*

$$P_\gamma(M,t) = \sum_{n=0}^{\infty} \gamma(M_n)\, t^n \ \in K(A_0)[[t]].$$

Theorem 11.1′. $P_\gamma(M,t) = \dfrac{f(t)}{\prod_{i=1}^{s}(1-t^{k_i})}$ with $f(t) \in K(A_0)[t]$; that is, $\prod_{i=1}^{s}(1-t^{k_i}) \cdot P_\gamma(M,t) = f(t)$ in $K(A_0)[[t]]$. (For every additive function λ with values in $\mathbf{Z}$ (or any abelian group), $P(M,t) = \sum \lambda(M_n)t^n$ is the image of $P_\gamma(M,t)$ under the coefficientwise map $\lambda_0 : K(A_0)[[t]] \to \mathbf{Z}[[t]]$, so Theorem 11.1 follows by applying λ_0 to f.)

Proof (the proof of 11.1 verbatim, with γ in place of λ; it only uses that λ is additive and takes values in an abelian group). Induction on s. If $s = 0$ then $A = A_0$; M is generated by finitely many homogeneous elements of degrees $\le r$, so $M_n = 0$ for $n > r$ and $P_\gamma(M,t)$ is a polynomial.

Let $s > 0$ and assume the result for rings generated by $s-1$ elements. Multiplication by x_s is an A_0-linear map $M_n \to M_{n+k_s}$; let $K_n = \mathrm{Ker}$ and $L_{n+k_s} = \mathrm{Coker}$, giving exact sequences of finitely generated A_0-modules

$$0 \to K_n \to M_n \xrightarrow{x_s} M_{n+k_s} \to L_{n+k_s} \to 0.$$

Put $K = \bigoplus K_n$, $L = \bigoplus L_n$ (with $L_n = M_n$ for $n < k_s$). K is a submodule and $L = M/x_sM$ a quotient of M, so they are finitely generated A-modules (as A is Noetherian), and both are annihilated by x_s; they are therefore finitely generated graded modules over the Noetherian graded ring $A/(x_s)$, which is generated over A_0 by the images of $x_1, \dots, x_{s-1}$. Since γ is additive, (2.11) gives $\gamma(K_n) - \gamma(M_n) + \gamma(M_{n+k_s}) - \gamma(L_{n+k_s}) = 0$. Multiply by t^{n+k_s} and sum over $n \ge 0$:

$$t^{k_s}P_\gamma(K,t) - t^{k_s}P_\gamma(M,t) + \big(P_\gamma(M,t) - g(t)\big) - \big(P_\gamma(L,t) - h(t)\big) = 0,$$

where $g = \sum_{j<k_s} \gamma(M_j)t^j$ and $h = \sum_{j<k_s} \gamma(L_j)t^j$ are polynomials. Hence $(1-t^{k_s})P_\gamma(M,t) = P_\gamma(L,t) - t^{k_s}P_\gamma(K,t) + g(t) - h(t)$. By the inductive hypothesis $P_\gamma(L,t)$ and $P_\gamma(K,t)$ have the form (polynomial)/$\prod_{i<s}(1-t^{k_i})$, and the result follows. □

Exercise 6

Problem. Let A be a ring (not necessarily Noetherian). Prove that

$$1 + \dim A \le \dim A[x] \le 1 + 2\dim A.$$

[Let $f : A \to A[x]$ be the embedding and consider the fiber of $f^* : \mathrm{Spec}(A[x]) \to \mathrm{Spec}(A)$ over a prime ideal $\mathfrak{p}$ of A. This fiber can be identified with the spectrum of $k \otimes_A A[x] \cong k[x]$, where k is the residue field at $\mathfrak{p}$ (Chapter 3, Exercise 21), and $\dim k[x] = 1$. Now use Exercise 7(ii) of Chapter 4.]

Solution. If $\dim A = \infty$ both inequalities say $\dim A[x] = \infty$, which follows from the lower bound below; so assume the dimension finite where needed.

Lower bound. Let $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ be a chain of primes of A. By Ch. 4, Ex. 7(ii) the ideals $\mathfrak{p}_i[x]$ are prime in $A[x]$, and they form a strictly increasing chain (compare constant polynomials).

Moreover $\mathfrak{p}_r[x] \subsetneq \mathfrak{p}_r[x] + (x)$, which is prime since $A[x]/(\mathfrak{p}_r[x] + (x)) \cong A/\mathfrak{p}_r$ is a domain. This is a chain of length $r+1$, so $\dim A[x] \geq 1 + \dim A$.

Upper bound. Let $Q_0 \subsetneq \cdots \subsetneq Q_m$ be a chain of primes of $A[x]$ and $\mathfrak{p}_i = Q_i \cap A$, so $\mathfrak{p}_0 \subseteq \cdots \subseteq \mathfrak{p}_m$. Fix a prime $\mathfrak{p}$ of A and consider the primes Q of $A[x]$ with $Q \cap A = \mathfrak{p}$. They contain $\mathfrak{p}[x]$, so correspond to primes of $(A/\mathfrak{p})[x]$ whose contraction to $A/\mathfrak{p}$ is 0, i.e. which do not meet $S = (A/\mathfrak{p}) \setminus \{0\}$; by 3.11(iv) these correspond, inclusion-preservingly, to primes of $S^{-1}((A/\mathfrak{p})[x]) = k(\mathfrak{p})[x]$ (this is the identification of the fibre of Ch. 3, Ex. 21(iv)). Now $k(\mathfrak{p})[x]$ is a principal ideal domain of dimension 1, so it has no chain of three distinct primes. Hence at most two of the Q_i have the same contraction $\mathfrak{p}$. The distinct values among $\mathfrak{p}_0 \subseteq \cdots \subseteq \mathfrak{p}_m$ form a strictly increasing chain of primes of A, so there are at most $\dim A + 1$ of them. Therefore $m + 1 \leq 2(\dim A + 1)$, i.e. $m \leq 1 + 2\dim A$. Hence $\dim A[x] \leq 1 + 2\dim A$. □

Exercise 7

Problem. Let A be a Noetherian ring. Then

$$\dim A[x] = 1 + \dim A,$$

and hence, by induction on n,

$$\dim A[x_1, \ldots, x_n] = n + \dim A.$$

[Let $\mathfrak{p}$ be a prime ideal of height m in A. Then there exist $a_1, \ldots, a_m \in \mathfrak{p}$ such that $\mathfrak{p}$ is a minimal prime ideal belonging to the ideal $\mathfrak{a} = (a_1, \ldots, a_m)$. By Exercise 7 of Chapter 4, $\mathfrak{p}[x]$ is a minimal prime ideal of $\mathfrak{a}[x]$ and therefore height $\mathfrak{p}[x] \leq m$. On the other hand, a chain of prime ideals $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_m = \mathfrak{p}$ gives rise to a chain $\mathfrak{p}_0[x] \subset \cdots \subset \mathfrak{p}_m[x] = \mathfrak{p}[x]$, hence height $\mathfrak{p}[x] \geq m$. Hence height $\mathfrak{p}[x]$ = height $\mathfrak{p}$. Now use the argument of Exercise 6.]

Solution. By Ex. 6, $\dim A[x] \geq 1 + \dim A$. For the reverse inequality it suffices to show:

$$(*) \qquad \text{for every prime } Q \text{ of } A[x] \text{ with } \mathfrak{p} = Q \cap A: \quad \operatorname{ht} Q \leq \operatorname{ht} \mathfrak{p} + 1,$$

because then $\dim A[x] = \sup_Q \operatorname{ht} Q \leq \sup_{\mathfrak{p}} \operatorname{ht} \mathfrak{p} + 1 = \dim A + 1$ (heights are finite by 11.12 since A and, by the Hilbert basis theorem, $A[x]$ are Noetherian).

Reduction to A local. Let $T = A \setminus \mathfrak{p}$. Every prime $Q' \subseteq Q$ of $A[x]$ has $Q' \cap A \subseteq \mathfrak{p}$, so it does not meet T; by 3.11 and 3.13, $\operatorname{ht} Q = \operatorname{ht}(T^{-1}Q)$ in $T^{-1}(A[x]) = A_{\mathfrak{p}}[x]$, and $T^{-1}Q \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$. So we may replace A by the Noetherian local ring $A_{\mathfrak{p}}$ and assume A local with maximal ideal $\mathfrak{p} = \mathfrak{m}$, $d = \operatorname{ht} \mathfrak{m} = \dim A$ and $Q \cap A = \mathfrak{m}$.

The case A local. By 11.13 there are $a_1, \ldots, a_d \in \mathfrak{m}$ with $\mathfrak{a} = (a_1, \ldots, a_d)$ $\mathfrak{m}$-primary. By Ch. 4, Ex. 7(iii), $\mathfrak{a}[x] = \mathfrak{a}A[x]$ is $\mathfrak{m}[x]$-primary, so $r(\mathfrak{a}A[x]) = \mathfrak{m}[x]$. Now $Q \supseteq \mathfrak{m}[x]$ and $Q/\mathfrak{m}[x]$ is a prime of $(A/\mathfrak{m})[x]$ whose contraction to $A/\mathfrak{m}$ is 0.

- If $Q = \mathfrak{m}[x]$, then Q is minimal over $\mathfrak{a}A[x]$ (its radical), an ideal generated by d elements, so $\operatorname{ht} Q \leq d$ by 11.16.
- If $Q \supsetneq \mathfrak{m}[x]$, then $Q/\mathfrak{m}[x]$ is a non-zero prime of the PID $(A/\mathfrak{m})[x]$, hence principal: $Q = \mathfrak{m}[x] + (f)$ for some $f \in A[x]$. The ideal $\mathfrak{b} = (a_1, \ldots, a_d, f)$ is contained in Q, so $r(\mathfrak{b}) \subseteq Q$; and $r(\mathfrak{b}) \supseteq r(\mathfrak{a}A[x]) = \mathfrak{m}[x]$ and $f \in r(\mathfrak{b})$, so $r(\mathfrak{b}) \supseteq Q$. Thus $r(\mathfrak{b}) = Q$ and Q is a minimal prime of $\mathfrak{b}$, which has $d+1$ generators; by 11.16, $\operatorname{ht} Q \leq d + 1$.

In either case $\operatorname{ht} Q \le d+1 = \operatorname{ht}\mathfrak{m}+1$, proving $(*)$. Hence $\dim A[x] = 1+\dim A$ (if $\dim A = \infty$, Ex. 6 gives $\dim A[x] = \infty$ as well).

Induction. $A[x_1,\dots,x_n] = A[x_1,\dots,x_{n-1}][x_n]$ and $A[x_1,\dots,x_{n-1}]$ is Noetherian (7.5), so $\dim A[x_1,\dots,x_n] = 1+\dim A[x_1,\dots,x_{n-1}] = n+\dim A$.

Remark (the hint's route). The argument above also shows $\operatorname{ht}\mathfrak{p}[x] = \operatorname{ht}\mathfrak{p}$: the inequality $\ge$ comes from the chain $\mathfrak{p}_0[x] \subset \cdots \subset \mathfrak{p}[x]$, and $\le$ from the case $Q = \mathfrak{m}[x]$ treated above (after localizing at $\mathfrak{p}$, where $\mathfrak{p}[x] \mapsto \mathfrak{m}[x]$). □

Notes

There are no numbered in-text exercises in Chapter 11, and none of the 7 end-of-chapter exercises is only exposition. In Exercise 5 the phrase "reformulate" leaves the form of the answer open; the statement given (Theorem 11.1′) is the natural universal version, from which 11.1 for any additive function follows. Exercise 4 is a verification of the assertions in its statement (it cites Ch. 7, Ex. 9 for Noetherianity; that exercise is taken as known).